



National Load Despatch Centre
राष्ट्रीय भार प्रेषण केंद्र
GRID CONTROLLER OF INDIA LIMITED
ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

(Government of India Enterprise/ भारत सरकार का उद्यम)
B-9, QUTUB INSTITUTIONAL AREA, KATWARIA SARAI, NEW DELHI -110016
बी-9, कुतुब इन्स्टीट्यूशनल एरिया, कटवारिया सराये, न्यू दिल्ली-110016

Ref: POSOCO/NLDC/SO/Daily PSP Report

दिनांक: 23.09.2024

To,

- कार्यकारी निदेशक, पू.क्षे.भा.प्रे.के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14 Golf Club Road, Tollygunge, Kolkata, 700033
- कार्यकारी निदेशक, ऊ.क्षे.भा.प्रे.के., 18/ए, शाहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Executive Director, NRLDC, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi - 110016
- कार्यकारी निदेशक, प.क्षे.भा.प्रे.के., एफ3, एम आई डी सी क्षेत्र, अंधेरी, मुंबई - 400093
Executive Director, WRLDC, F-3, M.I.D.C. Area, Marol, Andheri (East), Mumbai-400093
- कार्यकारी निदेशक, ऊ.पू.क्षे.भा.प्रे.के., डोंगतेह, लोअर नोंग्रह, लापलंग, शिलोंग - 793006
Executive Director, NERLDC, Dongteih, Lower Nongrah, Lapalang, Shillong - 793006, Meghalaya
- कार्यकारी निदेशक, द.क्षे.भा.प्रे.के., 29, रेस कोर्स क्रॉस रोड, बंगलुरु - 560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009

Sub: Daily PSP Report for the date 22.09.2024.

महोदय/Dear Sir,

आईईजीसी-2023 की धारा स.-5.5.1 के प्रावधान के अनुसार, दिनांक 22-सितंबर-2024 की अखिल भारतीय प्रणाली की दैनिक ग्रिड निष्पादन रिपोर्ट रांभांप्रेके की वेबसाइट पर उपलब्ध है।

As per article 5.5.1 of the Indian Electricity Grid Code, the daily report pertaining power supply position of All India Power System for the date 22.09.2024, is available at the NLDC website.

धन्यवाद,
Thanks

Report for previous day

Date of Reporting: 23-Sep-2024

A. Power Supply Position at All India and Regional level

	NR	WR	SR	ER	NER	TOTAL
Demand Met during Evening Peak hrs(MW) (at 20:00 hrs; from RLDCs)	71507	59422	44070	29415	3413	207827
Peak Shortage (MW)	185	0	0	0	60	245
Energy Met (MU)	1625	1455	1140	636	74	4930
Hydro Gen (MU)	316	118	143	95	27	699
Wind Gen (MU)	14	66	160	-	-	241
Solar Gen (MU)*	166.49	87.46	118.84	3.18	1.24	377
Energy Shortage (MU)	0.96	0.17	0.00	0.70	0.11	1.94
Maximum Demand Met During the Day (MW) (From NLDC SCADA)	72429	63935	53482	30799	3531	216244
Time Of Maximum Demand Met	22:12	14:31	09:33	23:01	18:16	14:52

B. Frequency Profile (%)

Region	FVI	< 49.7	49.7 - 49.8	49.8 - 49.9	< 49.9	49.9 - 50.05	> 50.05
All India	0.071	0.28	3.60	13.67	17.54	73.63	8.83

C. Power Supply Position in States

Region	States	Max.Demand Met during the day (MW)	Shortage during maximum Demand (MW)	Energy Met (MU)	Drawal Schedule (MU)	OD(+)/UD(-) (MU)	Max OD (MW)	Energy Shortage (MU)
NR	Punjab	15242	0	300.3	178.3	-0.8	526	0.00
	Haryana	10016	0	223.7	163.3	-1.2	275	0.00
	Rajasthan	14435	0	325.9	113.8	-0.2	740	0.00
	Delhi	5709	0	117.8	108.6	-0.4	250	0.00
	UP	26263	0	508.7	210.1	-1.3	944	0.00
	Uttarakhand	2248	0	48.5	22.7	0.2	151	0.03
	HP	1656	0	35.7	5.4	-0.8	76	0.00
	J&K(UT) & Ladakh(UT)	2588	214	53.7	31.4	4.7	648	0.93
	Chandigarh	322	0	6.2	6.2	0.0	34	0.00
WR	Railways_NR ISTS	203	0	4.3	3.8	0.5	37	0.00
	Chhattisgarh	5601	0	125.1	76.0	-1.8	352	0.00
	Gujarat	22410	0	471.4	184.8	-1.7	670	0.00
	MP	12139	0	264.1	132.9	-3.0	435	0.17
	Maharashtra	23180	0	527.0	170.8	3.8	854	0.00
	Goa	623	0	13.4	10.7	2.3	46	0.00
	DNHDDPDCL	1289	0	29.7	29.8	-0.1	239	0.00
	AMNSIL	552	0	11.4	2.6	0.3	192	0.00
	BALCO	527	0	12.6	12.6	0.0	4	0.00
SR	Andhra Pradesh	11154	0	225.8	75.3	-1.8	566	0.00
	Telangana	11319	0	231.8	89.8	-1.2	522	0.00
	Karnataka	13811	0	239.3	78.3	0.4	750	0.00
	Kerala	4012	0	79.0	51.2	0.6	241	0.00
	Tamil Nadu	15816	0	354.6	125.2	-2.8	812	0.00
	Puducherry	454	0	9.9	9.4	-0.1	55	0.00
ER	Bihar	7845	0	163.5	155.3	-4.3	230	0.70
	DVC	3442	0	73.6	-49.6	-2.0	177	0.00
	Jharkhand	2175	0	45.5	34.8	0.5	119	0.00
	Odisha	5903	0	124.6	25.2	-0.7	364	0.00
	West Bengal	11845	0	227.9	85.0	-1.5	323	0.00
	Sikkim	67	0	1.0	1.1	-0.1	13	0.00
	Railways_ER ISTS	22	0	0.2	0.1	0.1	4	0.00
NER	Arunachal Pradesh	205	0	3.5	3.1	0.1	39	0.00
	Assam	2399	0	52.2	43.7	1.5	7	0.11
	Manipur	212	0	2.8	2.7	0.1	42	0.00
	Meghalaya	261	0	4.3	1.3	-0.2	83	0.00
	Mizoram	118	0	2.2	0.5	-0.1	19	0.00
	Nagaland	184	0	3.2	3.0	0.0	30	0.00
	Tripura	350	0	5.8	5.1	0.3	68	0.00

D. Transnational Exchanges (MU) - Import(+ve)/Export(-ve)

	Bhutan	Nepal	Bangladesh	Godda -> Bangladesh
Actual (MU)	39.2	11.2	-12.3	-24.9
Day Peak (MW)	1943.0	781.5	-541.0	-1122.3

E. Import/Export by Regions (in MU) - Import(+ve)/Export(-ve); OD(+)/UD(-)

	NR	WR	SR	ER	NER	TOTAL
Schedule(MU)	357.5	-254.2	-19.7	-95.3	11.7	0.0
Actual(MU)	347.2	-234.8	-29.4	-100.1	12.5	-4.6
O/D/U/D(MU)	-10.3	19.5	-9.7	-4.8	0.8	-4.6

F. Generation Outage(MW)

	NR	WR	SR	ER	NER	TOTAL	% Share
Central Sector	4941	13678	4378	5716	259	28971	54
State Sector	5621	9448	7812	1994	155	25029	46
Total	10561	23125	12190	7710	414	54000	100

G. Sourcewise generation (Gross) (MU)

	NR	WR	SR	ER	NER	All India	% Share
Coal	797	1436	654	695	16	3597	68
Lignite	27	14	59	0	0	100	2
Hydro	316	118	143	95	27	699	13
Nuclear	30	53	67	0	0	150	3
Gas, Naptha & Diesel	13	39	5	0	23	80	2
RES (Wind, Solar, Biomass & Others)	186	155	307	6	1	655	12
Total	1368	1815	1235	795	67	5282	100

Share of RES in total generation (%)	13.62	8.56	24.83	0.71	1.84	12.41
Share of Non-fossil fuel (Hydro,Nuclear and RES) in total generation(%)	38.87	18.01	41.86	12.62	41.51	28.48

H. All India Demand Diversity Factor

Based on Regional Max Demands	1.036
Based on State Max Demands	1.094

I. All India Peak Demand and shortage at Solar and Non-Solar Hour

	Max Demand Met(MW)	Time	Shortage(MW)
Solar hr	216244	14:52	0
Non-Solar hr	209979	22:25	356

Diversity factor = Sum of regional or state maximum demands / All India maximum demand

**Note: All generation MU figures are gross

***Godda (Jharkhand) -> Bangladesh power exchange is through the radial connection (isolated from Indian Grid)

Solar Hours -> 06:00 to 18:00hrs and rest are Non-Solar Hours

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

Executive Director-NLDC

INTER-REGIONAL EXCHANGES

Import=(+ve) /Export =(-ve) for NET (MU)

Date of Reporting: 23-Sep-2024

Sl No	Voltage Level	Line Details	No. of Circuit	Max Import (MW)	Max Export (MW)	Import (MU)	Export (MU)	NET (MU)
Import/Export of ER (With NR)								
1	HVDC	ALIPURDUAR-AGRA	2	0	350	0.0	8.4	-8.4
2	HVDC	PUSAULI B/B	-	0	112	0.0	2.6	-2.6
3	765 kV	GAYA-VARANASI	2	242	602	0.0	2.8	-2.8
4	765 kV	SASARAM-FATEHPUR	1	155	274	0.0	2.0	-2.0
5	765 kV	GAYA-BALIA	1	0	807	0.0	15.5	-15.5
6	400 kV	PUSAULI-VARANASI	1	0	137	0.0	2.5	-2.5
7	400 kV	PUSAULI -ALLAHABAD	1	26	27	0.0	2.6	-2.6
8	400 kV	MUZAFFARPUR-GORAKHPUR	2	0	582	0.0	8.7	-8.7
9	400 kV	PATNA-BALIA	2	0	852	0.0	16.8	-16.8
10	400 kV	NAUBATPUR-BALIA	2	0	257	0.0	4.8	-4.8
11	400 kV	BIHARSHARIFF-BALIA	2	22	159	0.0	2.7	-2.7
12	400 kV	MOTIHARI-GORAKHPUR	2	0	390	0.0	7.0	-7.0
13	400 kV	BIHARSHARIFF-VARANASI	2	0	262	0.0	0.0	0.0
14	220 kV	SAHUPURI-KARAMNASA	1	0	143	0.0	0.6	-0.6
15	132 kV	NAGAR UNTARI-RIHAND	1	0	0	0.1	0.0	0.1
16	132 kV	GARWAH-RIHAND	1	30	0	0.7	0.0	0.7
17	132 kV	KARMANASA-SAHUPURI	1	0	0	0.0	0.0	0.0
18	132 kV	KARMANASA-CHANDAULI	1	0	0	0.0	0.0	0.0
ER-NR						0.8	76.9	-76.2
Import/Export of ER (With WR)								
1	765 kV	JHARSUGUDA-DHARAMJAIGARH	4	108	1113	0.0	12.8	-12.8
2	765 kV	NEW RANCHI-DHARAMJAIGARH	2	2656	0	37.9	0.0	37.9
3	765 kV	JHARSUGUDA-DURG	2	0	741	0.0	9.9	-9.9
4	400 kV	JHARSUGUDA-RAIGARH	4	0	828	0.0	12.8	-12.8
5	400 kV	RANCHI-SIPAT	2	539	81	6.1	0.0	6.1
6	220 kV	BUDHIPADAR-RAIGARH	1	0	140	0.0	2.2	-2.2
7	220 kV	BUDHIPADAR-KORBA	2	0	199	0.0	2.9	-2.9
ER-WR						44.0	40.6	3.4
Import/Export of ER (With SR)								
1	HVDC	JEYPORE-GAZUWAKA B/B	2	0	208	0.0	4.9	-4.9
2	HVDC	TALCHER-KOLAR BIPOLE	2	0	2	0.0	0.0	0.0
3	765 kV	ANGUL-SRIKAKULAM	2	0	1900	0.0	30.3	-30.3
4	400 kV	TALCHER-I/C	2	1396	0	31.8	0.0	31.8
5	220 kV	BALIMELA-UPPER-SILERRU	1	0	0	0.0	0.0	0.0
ER-SR						0.0	35.2	-35.2
Import/Export of ER (With NER)								
1	400 kV	BINAGURI-BONGAIGAON	2	0	436	0.0	7.5	-7.5
2	400 kV	ALIPURDUAR-BONGAIGAON	2	0	714	0.0	12.5	-12.5
3	220 kV	ALIPURDUAR-SALAKATI	2	0	138	0.0	2.7	-2.7
ER-NER						0.0	22.7	-22.7
Import/Export of NER (With NR)								
1	HVDC	BISWANATH CHARIALI-AGRA	2	0	502	0.0	12.0	-12.0
NER-NR						0.0	12.0	-12.0
Import/Export of WR (With NR)								
1	HVDC	CHAMPA-KURUKSHETRA	2	1	1	0.0	0.0	0.0
2	HVDC	VINDHYACHAL B/B	-	0	497	0.0	12.0	-12.0
3	HVDC	MUNDRA-MOHINDERGARH	2	0	2353	0.0	60.4	-60.4
4	765 kV	GWALIOR-AGRA	2	0	1858	0.2	31.3	-31.1
5	765 kV	GWALIOR-PHAGI	2	101	1997	0.1	32.2	-32.1
6	765 kV	JABALPUR-ORAI	2	0	1405	0.0	44.0	-44.0
7	765 kV	GWALIOR-ORAI	1	718	0	10.8	0.0	10.8
8	765 kV	SATNA-ORAI	1	0	834	0.0	20.0	-20.0
9	765 kV	BANASKANTHA-CHITORGARH	2	2024	52	19.7	0.0	19.7
10	765 kV	VINDHYACHAL-VARANASI	2	0	4138	0.0	72.9	-72.9
11	400 kV	ZERDA-KANKROLI	1	402	65	3.4	0.2	3.2
12	400 kV	ZERDA -BHINMAL	1	0	213	0.0	2.9	-2.9
13	400 kV	VINDHYACHAL -RIHAND	1	0	1	0.0	0.0	0.0
14	400 kV	RAPP-SHUJALPUR	2	220	626	0.5	8.0	-7.5
15	400 kV	NEEMUCH-Chittorgarh	2	13	729	0.0	10.3	-10.3
16	220 kV	BHANPURA-RANPUR	1	0	78	0.0	1.1	-1.1
17	220 kV	BHANPURA-MORAK	1	0	30	0.0	2.4	-2.4
18	220 kV	MEHGAON-AURAIYA	1	72	30	0.3	0.2	0.1
19	220 kV	MALANPUR-AURAIYA	1	44	49	0.1	0.4	-0.3
20	132 kV	GWALIOR-SAWAI MADHOPUR	1	0	0	0.0	0.0	0.0
21	132 kV	RAJGHAT-LALITPUR	2	0	0	0.0	0.0	0.0
WR-NR						35.0	298.3	-263.3
Import/Export of WR (With SR)								
1	HVDC	BHADRAWATI B/B	-	0	507	0.0	11.9	-11.9
2	HVDC	RAIGARH-PUGALUR	2	0	605	0.0	14.6	-14.6
3	765 kV	SOLAPUR-RAICHUR	2	1788	0	26.6	0.0	26.6
4	765 kV	WARDHA-NIZAMABAD	2	802	1117	4.6	7.4	-2.8
5	765 kV	WARORA-WARANGAL(NEW)	2	728	1163	3.6	8.3	-4.7
6	400 kV	KOLHAPUR-KUDGI	2	1958	0	35.8	0.0	35.8
7	220 kV	KOLHAPUR-CHIKODI	2	0	0	0.0	0.0	0.0
8	220 kV	PONDA-AMBEWADI	1	67	0	1.2	0.0	1.2
9	220 kV	XELDEM-AMBEWADI	1	0	110	2.2	0.0	2.2
WR-SR						74.1	42.3	31.8

INTERNATIONAL EXCHANGES					Import(+ve)/Export(-ve)		
State	Region	Line Name	Max (MW)	Min (MW)	Avg (MW)	Energy Exchange (MU)	
BHUTAN	ER	400kV MANGDECHHU-ALIPURDUAR 1,2&3 i.e. ALIPURDUAR RECEIPT (from MANGDECHU HEP 4*180MW)	696	562	595	14.27	
	ER	400kV TALA-BINAGURI 1,2,4 (& 400kV MALBASE - BINAGURI) i.e. BINAGURI RECEIPT (from TALA HEP 6*170MW)	917	811	837	20.09	
	ER	220kV CHUKHA-BIRPARA 1&2 (& 220kV MALBASE - BIRPARA) i.e. BIRPARA RECEIPT (from CHUKHA HEP 4*84MW)	167	69	117	2.82	
	NER	132kV GELEPHU-SALAKATI	33	0	25	0.60	
	NER	132kV MOTANGA-RANGIA	68	0	58	1.38	
NEPAL	NR	NEPAL IMPORT (FROM UP)	-38	0	0	0.00	
	NR	132kV MAHENDRANAGAR-TANAKPUR(NHPC)	66	0	40	0.95	
	ER	NEPAL IMPORT (FROM BIHAR)	41	6	41	1.82	
	ER	400kV DHALKEBAR-MUZAFFARPUR 1&2	674	122	351	8.42	
BANGLADESH	ER	BHERAMARA B/B HVDC (B'DESH)	-459	-438	-448	-10.74	
	ER (Isolated from Indian Grid)	400kV GODDA_TPS-RAHANPUR (B'DESH) D/C	-1122	-993	-1039	-24.93	
	NER	132kV COMILLA-SURAJMANI NAGAR 1&2	-82	0	-67	-1.60	

Export From India (in MU)

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Nepal	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02
Bangladesh	11.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.06
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Export	11.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.08

Import by India(in MU)

Country	GNA (ISGA/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	36.97	0.00	1.30	0.00	0.00	0.00	0.00	0.00	38.27
Nepal	7.36	0.00	5.27	0.00	0.00	0.00	0.00	0.00	12.63
Bangladesh	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Import	44.33	0.00	6.57	0.00	0.00	0.00	0.00	0.00	50.90

Net from India(in MU)

-ve : Export / +ve : Import

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	36.97	0.00	1.30	0.00	0.00	0.00	0.00	0.00	38.27
Nepal	7.34	0.00	5.27	0.00	0.00	0.00	0.00	0.00	12.61
Bangladesh	-11.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-11.06
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Net	33.25	0.00	6.57	0.00	0.00	0.00	0.00	0.00	39.82